Aryan Parmar

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Education

National Institute of Technology Surat (SVNIT Surat) — Surat, Gujarat, India
Bachelor of Technology in Electronics and Communication Engineering; CGPA: 7.37 / 10 — *2023 – Present*

GSHSEB Board — Gujarat, India
Higher Secondary Education (Class XII); Percentage: 77% — *2023*

GSEB Board — Gujarat, India
Secondary Education (Class X); Percentage: 76% — *2021*

Projects

QueueForge – Distributed Background Job Processing System | Node.js, Redis, React.js, REST API — 2026
- Architected and delivered a highly scalable distributed job processing system from scratch using Node.js and Redis, building core queue internals without relying on BullMQ or Agenda.
- Designed a four-process architecture (Producer, Worker, Scheduler, Dashboard) supporting concurrent multi-worker ownership, priority queues, delayed jobs, exponential backoff retries, and dead letter queues to keep the system stable under failure.
- Built and tested 8+ REST APIs over Redis lists, hashes, and sorted sets, validating the system against 1,000+ simulated jobs with real-time status tracking and telemetry.
- Developed a React operations dashboard with Recharts to surface throughput, worker activity, failure, and queue-depth metrics for operational visibility.

Real-Time Traffic Violation Detection System | Python, YOLOv8, OpenCV, EasyOCR — 2026
- Designed and implemented a dual-model detection pipeline using YOLOv8, OpenCV, and EasyOCR to identify helmet violations, triple riding, and license plates in real time.
- Built a test and evaluation framework across 1,000+ annotated frames to measure detection accuracy, automatically logging violations with visual evidence for audit.

Fake Social Media Account Detection System | Python, FastAPI, BERT, Random Forest, Streamlit — 2026
- Architected a fake-account detection service combining Random Forest and BERT models, integrating behavioral features with NLP-based profile analysis for higher classification accuracy.
- Engineered 15+ linguistic features using TextStat and developed a FastAPI backend with a Streamlit dashboard, enabling real-time predictions with measurable performance metrics.

Helper Buddy – Cleaning Service Booking Platform | Next.js, PostgreSQL, Prisma, Zustand — 2025
- Collaborated within a team to design and build a full-stack, scalable service booking platform using Next.js, Prisma, PostgreSQL, and Zustand across 10+ responsive pages.
- Modeled 8+ relational entities and implemented global state management, shipping the platform to production as part of the Google Winter of Code (GWOC) program.

Technical Skills

Programming Languages: C++, JavaScript, Python, SQL
Frameworks & Libraries: Node.js, Express.js, Next.js, React.js, Socket.IO, FastAPI, Tailwind CSS
Databases: MongoDB, MySQL, Redis
ORMs: Prisma, Mongoose
Core Concepts: Data Structures and Algorithms, REST API Design, Scalable System Design, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks
Testing & Tooling: Unit & Integration Testing, Test Automation, Postman, Git, GitHub, VS Code, Vercel, Docker

Achievements

Ranked in the **Top 10 out of 100+ teams** at the Echelon Hackathon organized at SVNIT Surat.

Solved **500+ problems** on LeetCode and Codeforces, strengthening core data structures, algorithms, and analytical problem-solving proficiency.

Positions of Responsibility

Google Developer Group (GDG), NIT Surat — Oct 2024 – May 2025
Junior Developer

Association for Computing Machinery (ACM) Club, NIT Surat — Aug 2024 – Mar 2025
Executive Member